

RAG Testing Document

Introduction

This PDF is created specifically for testing a Retrieval-Augmented Generation (RAG) chatbot. The chatbot should retrieve relevant information from this document before answering questions about it.

Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, perception, language understanding, and decision making.

Machine Learning

Machine Learning (ML) is a subset of AI in which computer systems learn patterns from data and use those patterns to make predictions or decisions. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with a language model. First, relevant documents or document chunks are retrieved from a knowledge base. The retrieved information is then provided to the language model as context so that it can generate an answer based on the available information.

Vector Database

A vector database stores numerical representations called embeddings. Embeddings represent the semantic meaning of text. During a RAG search, the user's question is converted into an embedding and compared with stored vectors to find semantically relevant document chunks.

FAISS

FAISS is a library designed for efficient similarity search over vectors. In a RAG application, FAISS can be used to store document embeddings and retrieve the chunks that are most similar to a user's query.

Deep Learning

Artificial Neural Networks

An Artificial Neural Network (ANN) is a machine learning model inspired by the structure of biological neural networks. An ANN commonly contains an input layer, one or more hidden layers, and an output layer. Neurons use weights, biases, and activation functions to transform information.

Transformers

The Transformer is a neural network architecture designed to process sequences using attention mechanisms. Self-attention allows the model to determine which parts of an input sequence are important in relation to other parts. Transformers are widely used in modern natural language processing systems.

Natural Language Processing

Natural Language Processing (NLP) is a field of AI concerned with enabling computers to understand, process, and generate human language. Applications include text classification, sentiment analysis, machine translation, question answering, and chatbots.

Testing Questions

- What is RAG?
- How does FAISS help in a RAG system?
- What is the difference between AI and Machine Learning?
- What is an embedding?
- What does self-attention do in a Transformer?
- What are the common layers of an ANN?
- What is NLP used for?